50

Answer-1

Electrostatic force,

$$F = \frac{k q_1 q_2}{r^2}$$

$$\Rightarrow r^2 = \frac{k q_1 q_2}{F}$$

$$\therefore r = \sqrt{\frac{9\times10^{9} \times 26\times10^{-6} \times 47\times10^{-6}}{5.7}}$$

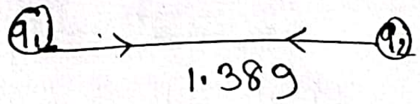
$$= 1.389 \text{ m}$$

given,

$q_1 = 26 \mu C$
$= 26\times10^{-6} C$

$q_2 = -47 \mu C$
$= -47\times10^{-6} C$

$F = 5.7 N$



And the force is attractive.

Ans

Answer-2

(a)

By newton's law,

$$F_1 = m_1 a_1 = 6.3\times10^{-7} \times 7$$
$$= 4.41\times10^{-6} N$$

given;

$a_1 = 7 ms^{-2}$

$a_2 = 9 ms^{-2}$

$m_1 = 6.3\times10^{-7} kg$

$$F_2 = m_2 a_2$$

$$\therefore m_2 = \frac{F_2}{a_2} = \frac{4.41\times10^{-6}}{9} = 4.9\times10^{-7} \text{ kg}$$

(b)

Applying coulomb's law,

$$\boxed{F = \frac{k q q}{d^2}}$$

$$q^2 = \frac{F d^2}{k}$$

$$\therefore q = \sqrt{\frac{4.41\times10^{-6} \times (3.2\times10^{-3})^2}{9\times10^{9}}} = 7.08\times10^{-11} C$$

$\boxed{q_1 = q_2 = q}$

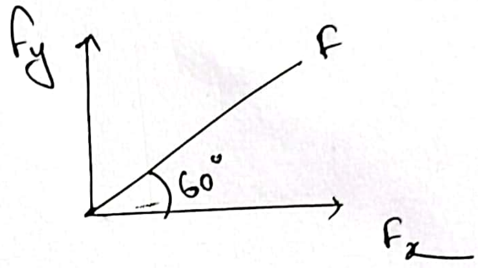
$\boxed{d = 3.2\times10^{-3} m}$

Answer – 03

(i)

$F_x = F\cos 60^\circ$

$F_y = F\sin 60^\circ$



Now,

$$F_x = F\cos 60^\circ$$

$$40 = F \times \frac{1}{2}$$

$$\therefore F = 80\,N$$

given,

$F_x = 40\,N$

Again,

$$F_y = F\sin 60^\circ$$

$$= 80 \times \frac{\sqrt{3}}{2} = 40\sqrt{3}\,N$$

(ii)

To find the angle between $\vec{F_1}, \vec{F_2}$ –

given,

$\vec{F_1} = 3\hat{i} - 4\hat{j}$

$\vec{F_2} = -2\hat{i} + 3\hat{k}$

$$\boxed{\vec{F_1} \cdot \vec{F_2} = |\vec{F_1}|\,|\vec{F_2}| \cos\theta}$$

$$\Rightarrow \cos\theta = \frac{\vec{F_1} \cdot \vec{F_2}}{|\vec{F_1}|\,|\vec{F_2}|}$$

$$= \frac{-6 - 0 + 0}{5 \cdot \sqrt{13}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{-6}{5\sqrt{13}}\right) = 109.44^\circ$$

Answer - 4

a) $\vec{A} = x\hat{i} + y\hat{j}$

$= -25\hat{i} + 40\hat{j}$

$\therefore |\vec{A}| = \sqrt{(-25)^2 + 40^2}$

$= 47.17\,m$

b) $\vec{A} \cdot x = \vec{A} \cdot \hat{i} = |\vec{A}||\hat{i}| \cos\theta$

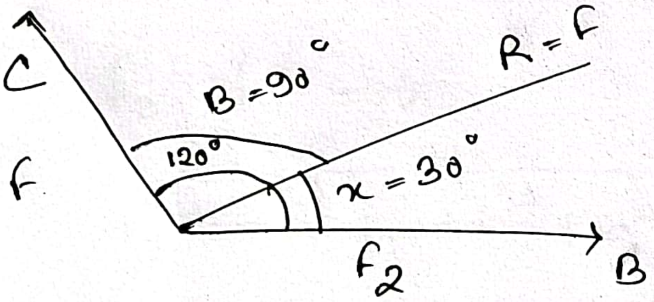
$\Rightarrow \cos\theta = \dfrac{\vec{A} \cdot \hat{i}}{|\vec{A}||\hat{i}|}$

$\therefore \theta = \cos^{-1}\left\{ \dfrac{(-25\hat{i} + 40\hat{j})\hat{i}}{\cancel{4} \cancel{\times} + 47.17 \times 1} \right\}$

$= 122^\circ$

Ans.

Answer - 5

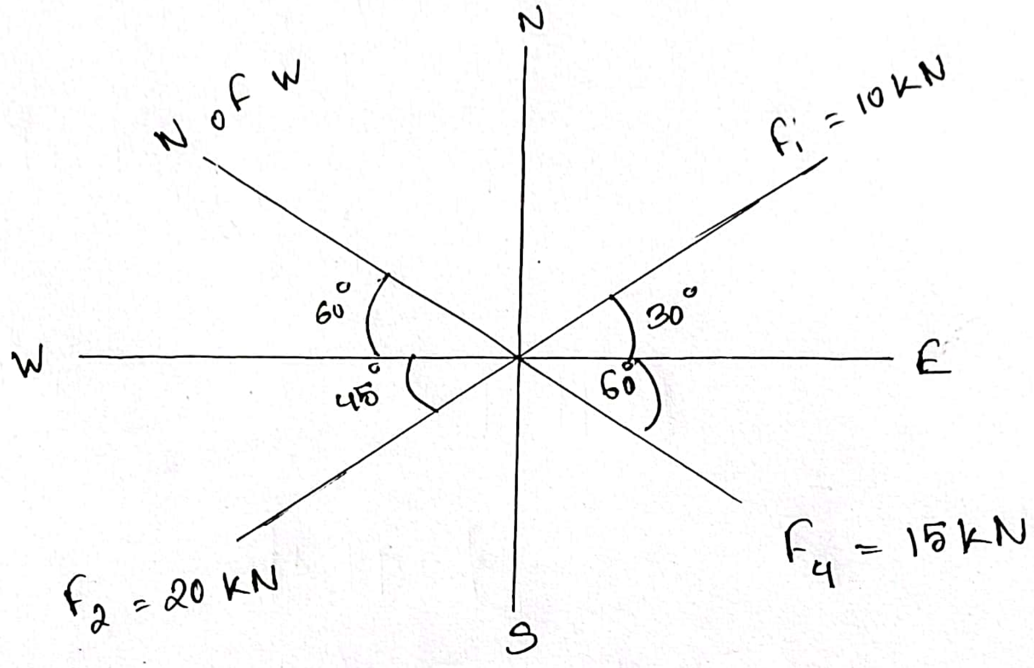


F_1 using the formula,

$F_1 = \dfrac{F \sin\beta}{\sin(\alpha + \beta)} \Rightarrow F = \dfrac{5 \sin(120^\circ)}{\sin 90^\circ} = 4.33\,N$

$F_2 = \dfrac{F \sin\alpha}{\sin(\alpha + \beta)} = \dfrac{4.3 \sin 30^\circ}{\sin 120^\circ} = 2.5\,N$

Answer – 6



$f_1, f_2 \longrightarrow$

$$F_{12} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 15°}$$

$$= \sqrt{10^2 + 20^2 + 2 \times 10 \times 20 \cos 15°}$$

$$= 29.77\ kN$$

$f_3, f_4 \longrightarrow$

$$f_{34} = \sqrt{f_3^2 + f_4^2 + 2f_3f_4\cos 180°} = 10\ kN$$

$$\theta_{net} = \tan^{-1}\left\{\frac{f_{34}\sin 82.5°}{f_{12} + f_{34}\cos 82.5°}\right\} = 17.73$$

$\therefore\ \theta_{net} = 17.73 + 37.5 = 55.23°$ is with $+x$ or clockwise.

Answer - 7

given,

$F = 2 \times 10^{-3}$ N

Let, $q_1 = q$ and $q_2 = 2q$

Let, r be distance between them,

$$F = \frac{k q_1 q_2}{r^2}$$

$$\Rightarrow 3 \times 10^{-3} = \frac{k \cdot 2q^2}{r^2}$$

$$\therefore kq^2 = 1.5 \times 10^{-3} \times r^2 \quad \text{---} (1)$$

when r changes to $(r + 0.1)$m, force is 5×10^{-4} N

$$F = \frac{kq^2 \cdot 2}{(r + 0.1)^2}$$

$$\Rightarrow \cancel{r + 0} \frac{F}{k} = \left(\frac{q}{r + 0.1}\right)^2$$

$$\Rightarrow \frac{5 \times 10^{-4}}{1.5 \times 10^{-3}}$$

$$\Rightarrow 2.5 \times 10^{-4} = \frac{1.5 \times 10^{-3} r^2}{(r + 0.1)^2}$$

$$\Rightarrow 0.48 = \frac{r}{r + 0.1}$$

$$\therefore r = 0.689 \text{ m}$$

from (1)

$$k \cdot q^2 = 1.5 \times 10^{-3} \times r^2$$

$$q = 2.8 \times 10^{-8} \text{ C}$$

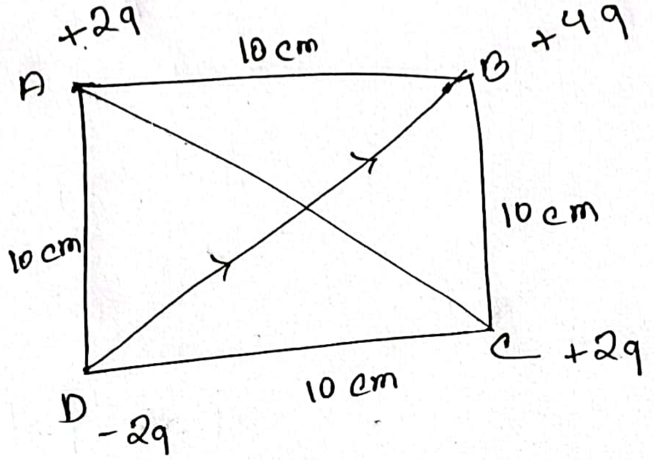
$$q_1 = q$$

$$q_2 = 2q = 5.6 \times 10^{-8}$$

Ans.

Answer - 8

(i)



(ii)

$$f_1 = \frac{k q_1 q_2}{r^2} = \frac{-2q^2 k}{r^2}$$

Diagonal of a square,

$$D = \sqrt{2} a$$

$$r = \frac{\sqrt{2} . a}{2} = 0.07 m$$

$$f_2 = \frac{k . 4q^2}{r^2}$$

$$f_3 = \frac{k . 2q^2}{r^2}$$

$$f_4 = \frac{k 2q^2}{r^2}$$

f_1 and f_3, $f_3 - f_1 = \frac{k . 2q^2}{r} - \frac{k . 2q^2}{r}$

$= 0N$

f_2 and f_4, $f_2 + f_2 = \frac{k . 6q^2}{r^2}$

$$f_{Net} = \sqrt{0^2 + f_{24}^2} = f_{24} = \frac{k . 6q^2}{r^2}$$

ABCD, $\cos\theta = \frac{CD}{BD}$

$$\theta = \cos^{-1}\left\{\frac{0.1}{0.1\sqrt{2}}\right\} = 45^\circ$$

$\therefore$ θ_{net} 45° along x axis.

Answer-10

Applying Coulomb's Law ;

$$F = \frac{kq_1q_2}{R^2} = \frac{kq^2}{r^2}$$

$$q = 2ne$$

1 mol Na = 23 gm

1 mol Na e^- available = Na

23g " e^- " 6.023×10^{23}

4g " e^- " $\frac{4 \times 6.023 \times 10^{23}}{23}$

$$n = 1.047 \times 10^{23}$$

$$\therefore q = 11 \times 0.047 \times 10^{23} \times 1.6 \times 10^{-19}$$

$$= 1.84 \times 10 \ 5C$$

from (1) $\rightarrow$ $F = \frac{kq^2}{R^2} = 9 \times 10^9 \times \frac{(1.84 \times 10^5)^2}{(2)}$

$$= 7.61 \times 10^{19} \ N$$

Answer-11

$$F_{net} = \vec{F_1} + \vec{F_2}$$

$$F_1 = \frac{kq_1q_2}{r^2} = \frac{k \times 25 \times (-12)}{2^2} = \frac{-300k}{4}$$

$$F_2 = \frac{kq_1q_3}{r^2} = \frac{k \times 25 \times (18)}{3^2} = \frac{450k}{9}$$

$$\therefore F = F_1 - F_2 = -2.25 \times 10^{-11} \ nN$$

$$\theta_{net} = \tan^{-1}\left\{\frac{F_2 \sin 180^\circ}{F_1 + F_2 \cos 180^\circ}\right\} = 0$$

Ans.

Answer-12

distance, $r = 5\,mm = 5\times10^{-9}\,m$

$G = 6.67\times10^{-11}\,m^3kg^{-1}s^{-2}$

mass $m_1 = m_2 = m = 1.672\times10^{-23}\,kg$

Electrostatic force,

$$f_E = \frac{k q_1 q_2}{r^2} = 9\times10^9 \times \frac{1.6\times10^{-19}\times1.6\times10^{-19}}{(5\times10^{-9})^2}$$

$$= 9.21\times10^{-12}$$

Gravitation force,

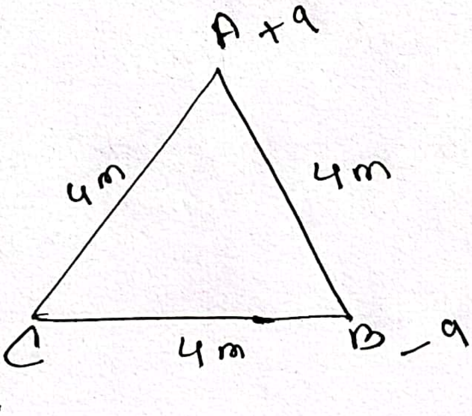
$$f_G = \frac{G m_1 m_2}{r^2} = 7.46\times10^{-48}\,N$$

$$\therefore \frac{f_E}{f_G} = \frac{9.21\times10^{-12}}{7.46\times10^{-48}}$$

$$\therefore f_E = 1.23\times10^{36} f_G$$

$$f_E \approx 10^{36} f_G$$

Answer-13



$q = 100\times10^{-6}\,C$

$r = 4\,m$

$$f_{AB} = \frac{k q_A q_B}{r^2} = 9\times10^9 \times \frac{100\times10^{-6}\times100\times10^{-6}}{4^2} = 5.675\,N$$

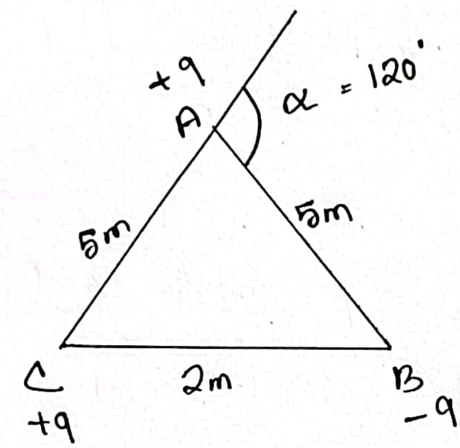
$$f_{CB} = \frac{k q_C q_B}{r^2} = 9\times10^9 \times \frac{100\times10^{-6}\times100\times10^{-6}}{4^2} = 5.625\,N$$

$$f = \sqrt{f_{AB}^2 + f_{CB}^2 + 2(f_{AB} - f_{CB}\cos60^\circ} = 9.74\,N$$

$$\tan\theta = \frac{f_{AB}\sin60^\circ}{f_{CB} + f_{AB}\cos60^\circ} \quad \therefore \theta = 30^\circ$$

Ans.

Answer – 14



Here,

$$q_A = q_B = q_C = 20\,nC$$
$$= 20 \times 10^{-19}\,C$$

$$r = r_{CA} = r_{BA} = 5\,m$$

$$r_{BC} = 2\,m$$

$$F_{CA} = k\frac{q_A q_C}{r^2}$$

$$= \frac{9\times10^{9} \times 20 \times 10^{-9} \times 20\times10^{-9}}{\cancel{1.44}\ 5^2}$$

$$= 1.44 \times 10^{-7}\,N$$

$$F_{BA} = k\frac{q_A q_B}{r^2}$$

$$= 1.44 \times 10^{-7}\,N$$

The magnitude,

$$F = \sqrt{F_{CA}^2 + F_{BA}^2 + 2F_{CA}F_{BA}\cos 120°}$$

$$= \sqrt{F^2 + F^2 - F.F}$$

$$= F = 1.44 \times 10^{-7}\,N$$

$$\theta = \tan^{-1}\frac{F_{CA}\sin 120°}{F_{BA} + F_{CA}\cos 120°} = 60°$$

Ans.

Answer – 15

$$F_1 = \frac{kq_0q_1}{r^2}$$

$$F_2 = \frac{kq_0q_2}{r^2}$$

$$q_1 = 3.2\times10^{-19}\,C$$
$$q_2 = 5.6\times10^{-19}\,C$$
$$q_0 = 1.6\times10^{-19}\,C$$

$$\therefore F_x = -F_1\cos\theta + F_2\cos\theta$$

$$= \frac{kq_0}{r^2}(-q_1 + q_2)\cos 30° = 1.19\times10^{-23}\,N$$

$$\therefore F_y = F_1 \sin\theta + F_2 \sin\theta$$

$$= \frac{kq_0}{r^2} \times \sin 30^\circ \left(3.2 \times 10^{-19} + 5.6 \times 10^{-19}\right)$$

$$= 2.53 \times 10^{-23}\ N$$

Magnitude net force,

$$\left|\vec{F}_{net}\right| = \sqrt{F_x^2 + F_y^2}$$

$$= 2.799 \times 10^{-23}\ N$$

(b)

$$\text{direction, } \theta = \tan^{-1}\left|\frac{F_y}{F_x}\right| = \boxed{64.73^\circ}$$

Ans

Answer - 16

The magnitude of the force is →

$$F = \frac{kq_1q_2}{r^2} = \frac{9 \times 10^9 \times \left(1.6 \times 10^{-19}\right)^2}{\left(3.8 \times 10^{-9}\right)^2} = 1.58\ N$$

direction,

$$\theta_{net} = \tan^{-1}\left\{\frac{F_2 \sin 60^\circ}{F_1 + F_2 \cos 60^\circ}\right\}$$

$$= 30^\circ$$

Ans.

Answer - 17

(a)

$$F_1 = \frac{k q_0 q_1}{r^2} = 0.45\ N$$

$$F_2 = \frac{k q_0 q_2}{r^2} = 0.45\ N$$

$$F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha}$$

$$= 0.78\ N$$

Answer - 18

$$F_{net} \Rightarrow F_1 + F_2 = F_3$$

$$0 = -\frac{k q_1 q}{r_1^2} + \frac{k q q_2}{r_2^2} - \frac{k q q_3}{r_3^2}$$

$$\Rightarrow 0 = -\frac{8}{8^2} + \frac{5}{14^2} - \frac{q_3}{26^2}$$

$$\therefore q_3 = -67.25\ C$$

Answer - 19

We know,

$$F = \frac{k q_1 q_2}{r^2} = \frac{kq^2}{r^2}$$

$$\therefore r = \sqrt{\frac{kq^2}{mg}}$$

$$= \sqrt{\frac{9\times10^9 \times (1.6\times10^{-19})^2}{1.67\times10^{-19} \times 9.8}}$$

$$= 0.119\ m$$

Ans.